Linear Regression

STAT 200 – Introductory Statistics

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Today's Objective

After today's lecture, you should:

- Understand the theory behind testing the relationship between two numeric variables.
- Estimate a value of the dependent variable (and provide a confidence interval).
- Predict a value of the dependent variable (and provide a prediction interval).

Example #1: Violent Crime Rate in 1990 vs. 2000

Example

Understanding the persistence of violent crime in cities can help determine which cities are being successful in reducing it and which cities are not. What is the relationship between the violent crime rate in 1990 and in 2000?

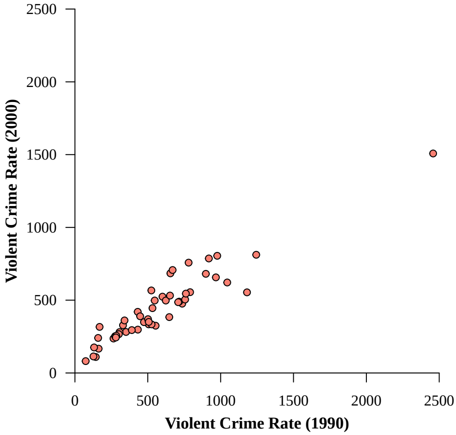
Note that we want to know *more* than just if there is a significant relationship.

We want to:

- Specify how much one influences the other.
- Estimate violent crime rates in 2000, given the value in 1990.
- Predict violent crime rates in 2000, given the value in 1990.

Example #1: Violent Crime Rate in 1990 vs. 2000

What is the relationship between these two variables?



The Theory

To summarize the relationship, we use a “line of best fit”:

$$\hat{y} = \beta_0 + \beta_1 x$$

However, this raises the following question: What do we mean by the “best” fit?

The Theory

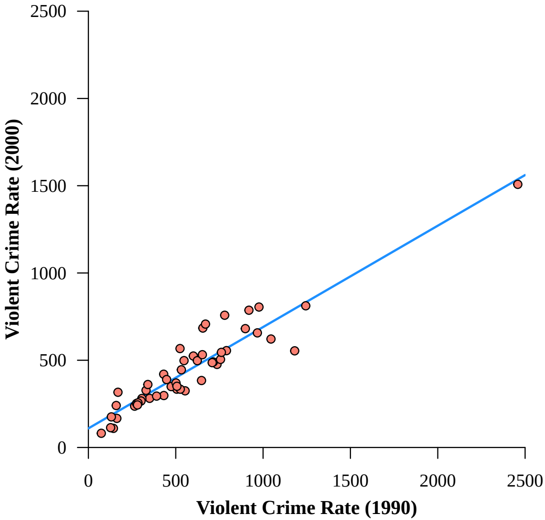
This is the most important question we can answer. Different answers lead to different fitting methods.

MATH/STAT 225 covers a few of the different methods. It also covers the easiest method in great detail.

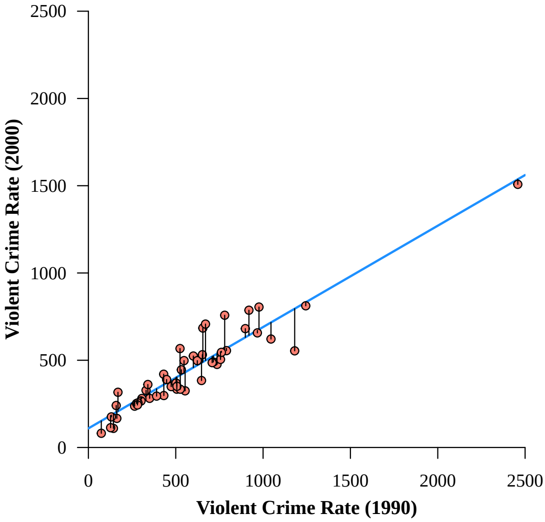
In this course, we will only look at the easiest method: ordinary least squares (OLS).

It is based on **minimizing the sum of the squared residual values**.

The Theory: The Line



The Theory: The Residuals



Recall that we are defining “best fit” as the line that minimizes the sum of the squared errors (residuals). The process requires differential calculus. If you’ve not had calculus, you can space out for a moment. Here’s a great use for it, try to follow along.

The Theory (Space Out)

The first step in minimization is to determine the objective (target) function that needs to be minimized.

$$\begin{aligned} Q &= \sum_i e_i^2 \\ &= \sum_i (y_i - \hat{y}_i)^2 \\ &= \sum_i (y_i - (\beta_0 + \beta_1 x_i))^2 \end{aligned}$$

This is our objective function. To minimize it, we take its derivative with respect to each parameter, set each equal to 0, and solve for the parameter.

The Theory (Space Out)

$$\frac{\partial}{\partial \beta_0} Q = \frac{\partial}{\partial \beta_0} \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$
$$= \sum_i 2(y_i - \beta_0 - \beta_1 x_i)(-1)$$

$$0 \overset{\text{set}}{=} -2 \left(\sum_i y_i - \sum_i b_0 - \sum_i b_1 x_i \right)$$
$$= n\overline{y} - nb_0 - n\overline{x}b_1$$

$$\Longrightarrow \qquad b_0 = \overline{y} - b_1 \overline{x}$$

Do not forget the definition of the sample mean:

$$\overline{x} = \frac{1}{n} \sum x_i \iff \sum x_i = n\overline{x}$$

The Theory (Space Out)

$$\frac{\partial}{\partial \beta_1} Q = \frac{\partial}{\partial \beta_1} \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$
$$= \sum_i 2(y_i - \beta_0 - \beta_1 x_i)(-x_i)$$

$$0 \overset{\text{set}}{=} -2 \left(\sum_i x_i y_i - \sum_i b_0 x_i - \sum_i b_1 x_i \right)$$
$$= \sum_i x_i y_i - (\overline{y} - b_1 \overline{x}) \sum_i x_i - \sum_i b_1 x_i^2$$
$$= \sum_i x_i y_i - (\overline{y} - b_1 \overline{x}) n \overline{x} - b_1 \sum_i x_i^2$$
$$= \sum_i x_i y_i - n \overline{x} \overline{y} - b_1 n \overline{x}^2) - b_1 \sum_i x_i^2$$

The Theory (Space Out)

$$
\begin{aligned}
0 &= \sum_i x_i y_i - n\overline{x}\overline{y} - b_1 n\overline{x}^2) - b_1 \sum_i x_i^2 \\
b_1 \sum_i x_i^2 - b_1 n\overline{x}^2 &= \sum_i x_i y_i - n\overline{x}\overline{y}
\end{aligned}
$$

$$
b_1 = \frac{\sum x_i y_i - n\overline{x}\overline{y}}{\sum x_i^2 - n\overline{x}^2}
$$

The Theory (Wake Up!)

Thus, our estimators of the y-intercept and slope (effect) are:

$$b_0 = \overline{y} - b_1\overline{x}$$
$$b_1 = \frac{\sum x_i y_i - n\overline{xy}}{\sum x_i^2 - n\overline{x}^2}$$

By the way, the second formula can also be written as:

$$b_1 = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})}{\sum(x_i - \overline{x})^2}$$
$$= r\left(\frac{s_y}{s_x}\right)$$

This gives some insight into the slope, the correlation, and the relationship between them.

Formulas for Linear Regression

To estimate the line of best fit, we have the formula below:

$$\hat{y} = b_0 + b_1 x$$

And we can calculate b_0 and b_1 with the formulas:

$$b_1 = r\left(\frac{s_y}{s_x}\right)$$

$$b_0 = \overline{y} - b_1\overline{x}$$

Remember, we have to find the slope, b_1, first before we can find the y-intercept, b_0.

Example #2: Linear Regression

Suppose we have two variables, x and y, with the following summary statistics:

	$\boldsymbol{x}$	$\boldsymbol{y}$
Mean	8	26
Std. Dev.	2	8
Correlation	0.75	

Find the line of best fit for predicting y using x.

Example #2: Linear Regression

The average STAT 200 student studies for 7 hours in the week before the midterm exam, with a variance of 9 hours squared, and sleeps for an average of 45 hours that week, with a variance of 144 hours squared. The correlation between the hours spent studying and the hours spent sleeping is $r = -0.60$.

Using the fact that Sam studied for 8 hours the week before the exam, predict how many hours he slept that week.

The Theory: `R` Help

Of course, for any real data set, it is unreasonable to do calculations by hand.

The calculations can be done in `R`, of course:

- The first step is to fit the model.
- The second step is to summarize the results.

We will return to this later in the course.